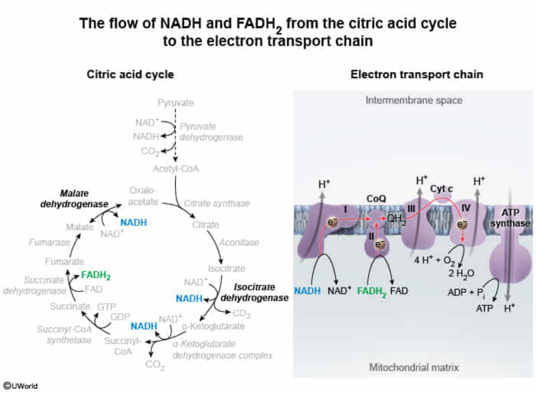
Which of the following is a citric acid cycle enzyme that produces a metabolite used directly in the electron transport chain?

I. Malate dehydrogenase
II. Glyceraldehyde 3-phosphate dehydrogenase
III. Aconitase
IV. Fumarase
V. Isocitrate dehydrogenase

A. I and V only [58%]
B. II and V only [6%]
C. III and IV only [13%]
D. I, II, and V only [22%]

Correct
59% answered correctly

Explanation:



The citric acid cycle (CAC) is an eight-enzyme metabolic cycle that uses NAD^+ and FAD to oxidize acetyl-CoA, producing CO_2 and the reduced molecules NADH and $FADH_2$. Three NADH and one $FADH_2$ molecules are produced by redox reactions per round of the CAC, and these reduced molecules subsequently enter Complexes I and II of the electron transport chain (ETC), where their oxidation back to NAD^+ and FAD is used to power ATP synthesis.

The enzymes that catalyze redox reactions (ie, oxidoreductases) are distinguished by their name "dehydrogenases," because the transfer of electrons is often accompanied by the transfer of hydrogen atoms as well. In the CAC, the oxidoreductases are **isocitrate dehydrogenase**, **α-ketoglutarate dehydrogenase**, **malate dehydrogenase**, and **succinate dehydrogenase** (also known as **Complex II of the ETC**). The first three enzymes listed reduce NAD^+ to NADH, and succinate dehydrogenase reduces FAD to $FADH_2$. Of these, only malate dehydrogenase and isocitrate dehydrogenase appear as options **(Numbers I and V)**.

(Number II) Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) is an oxidoreductase enzyme that reduces NAD^+ to NADH; however, GAPDH is an enzyme of glycolysis, *not* of the CAC. In addition, cytosolic NADH produced during glycolysis does not enter the ETC directly; the high-energy electrons must instead pass through the malate–aspartate shuttle or the glycerol

3-phosphate dehydrogenase shuttle to enter the ETC.

(Numbers III and IV) Although aconitase and fumarase are enzymes of the CAC, neither of these enzymes reduce NAD^+ to NADH or FAD to $FADH_2$.

Educational objective:

The citric acid cycle oxidizes acetyl-CoA to carbon dioxide while reducing NAD^+ to NADH and FAD to $FADH_2$. The resulting high-energy electrons can subsequently enter the electron transport chain. The enzymes that reduce these molecules are isocitrate dehydrogenase, α-ketoglutarate dehydrogenase, and malate dehydrogenase, which reduce NAD^+ to NADH, as well as succinate dehydrogenase, which reduces FAD to $FADH_2$.

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